Update to EEGS TD circle symbology. TD circle is now the correct size of 35mr. As the arc decreases, dots are shown. Also, an aspect caret is on the perimeter of the circle similar to the missile reticles. In-range cue moved to ~4 o'clock position and its size was adjusted.

Multiple fixes to RADAR AUTO mode for the HUD altitude scale. The 1,000-1,500 foot area is now correctly proportioned and has correct tick sizing and the thermometer tape scales for that region of the display properly (instead of reading linear with the rest of the scale below it). ALOW "T" on the thermometer tape better proportioned. AUTO scale now scrolls down the HUD when the gear is down with the FPM and airspeed scales. Erroneous "R" marking removed from thermometer tape. AL digital readout moved back below the altitude tape for AUTO mode in low case with thermometer shown (instead of overwriting the tape). In general, should look more like the footage you see in the LANTIRN HUD shots that show this altitude tape style now. Also, both airspeed and altitude tapes have the appropriate number of tic marks. The HUD VVI is now visible when using the ILS with the scales switch to VV/VAH. A-A TD box is now visible when using the ILS if a target is bugged.

Major overhaul to the pitch ladder drawing code. Most of this is aimed at making it easier to understand and maintain. However, it includes revised code for getting the ladder properly centered on the FPM regardless of drift mode selection, plus code to ensure the horizon line is always drawn on the horizon regardless of beta angle, better negative pitch line rendition, and improvements to the ghost horizon line. Also, the heading scale in the low position is pushed down the HUD by the FPM with Drift C/O engaged as well as in NORM mode now.

AMRAAM Dynamic Launch Zone (DLZ) Updates

The appearance and terminology of the AMRAAM DLZ has changed if the M2-style AMRAAM DLZ option is selected. The following is a summary:

R_{aero} – Raero (Range Aerodynamic) (small triangle) represents the maximum kinematic range of the AMRAAM and is the longest possible shot that a pilot can take and have a chance of hitting the target. It assumes the target will not maneuver and that the missile is perfect. It currently does not take optimum loft conditions into account (which should further increase range).

Ropt – Ropt (Range Optimum) (small circle) represents the maximum launch range with some buffer to offset any target turns. It is 85% of Raero. It currently does not assume the shooter will achieve optimum steering and pitch.

Rpi – Rpi (Range Probability of Intercept) is 75% of Raero and is a range that has a higher probability of hitting the target.

Rtr – Rtr (Range Turn and Run) replaces Rmax2 and represents the maximum range shot assuming that the target performs a turn-and-run maneuver at launch. Technical calculations are described on pg 106.

Rmin – Rmin replaces Rmin1 and Rmin2 and represents the minimum launch range. It is based on Rmin1 calculations.

AMRAAM Loft Solution Cue – A loft cue was added to aid the pilot achieve a lofted shot. It is above the Raero caret on the HUD and MFD and consists of a one or two-digit number representing the appropriate climb angle. When the DLZ target range cue reaches Raero the loft angle provides the conditions required to achieve an Raero shot (target range between Raero and Ropt) or an Ropt shot (target range between Ropt and Rpi). For either shot, center the Attack Steering Cue (ASC) in azimuth relative to the Allowable Steering Error Circle (ASEC) and execute a smooth pull-up to the indicated loft angle placing the ASC in the center of the ASEC. The loft solution cue counts down (every 5 degrees) from the total required angle as the pilot pitches the aircraft up to the MMC-specified loft angle. When the angle is reached, the cue will read “0” (zero) degrees to indicate that the pitch has been achieved. The loft cue is removed when the target range is less than Rpi. The loft cue does not take into account aircraft AOA while the display of the ASC does. This can result in a situation where the ASC is in the center of the ASEC (dot in the hole), but the loft solution cue is indicating a required loft angle. The greater the AOA is the greater the difference between the

positioning of the ASC and the display of the required loft angle. The pilot should center the dot in the hole in this case. In addition there is a two digit value above the target closure value which represents a Digital Maneuvering Cue (DMC). This is an angle value sort of representing how much turn a target would have to make to significantly degrade probability of achieving a missile intercept.

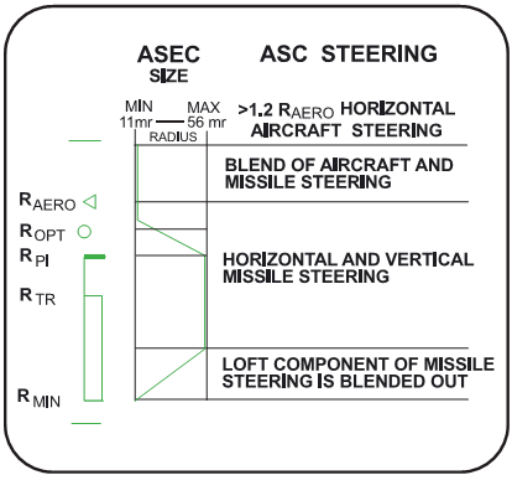
AIM-120 Boresight Reticle – When the selected AIM-120 sighting option is Bore, or the selected sighting is Slave and there is no bugged target, the AIM-120 boresight reticle (262mr diameter circle) is displayed on the HUD (the Bore reticle is not displayed on the MFD).

AIM-120 Missile Diamond – The missile diamond flashes when the target is in range (Raero or less).

Allowable Steering Error Circle (ASEC) – The ASEC is a variable diameter circle displayed on the HUD and MFD when an AIM-120 is the selected weapon, sighting option is Slave, and a bugged target exists. The weapon status must be RDY or SIM (Master ARM in ARM or SIM) in order for the ASEC/ASC and DLZ to appear. If there is no bugged target, the AIM-120 Boresight reticle will be displayed. The ASEC is an aid for positioning the attack steering cue in order to take the best shot possible based on the steering. At target ranges from outside Raero to Ropt, the ASEC is its smallest size, 11mr radius. At Ropt, the ASEC begins to grow in size until target range reaches Rpi where it reaches its maximum size (the ASEC represents 45 degrees of allowable steering error at Rpi). From Rpi to mid-point Rtr, the ASEC remains its largest size, at which point it begins to shrink again until it reaches minimum size at Rmin. The ASEC flashes when the target range is within the maneuver zone. The ASEC's variable radius varies from 11mr to 56mr with a slaved target. For a bore shot, the radius is static at 131mr (262mr diameter). The ASEC on the MFD is identical in function.

Attack Steering Cue (ASC) – The ASC (8 mr diameter circle in the HUD, 10 pixel radius circle on the MFD) provides several types of steering: horizontal aircraft steering, a blend of aircraft and missile steering, optimal missile steering (with horizontal and vertical missile steering), or Rmin steering (shortest LOS to the target). The type of steering provided is a function of range to the target. Horizontal aircraft steering is provided against targets beyond 1.2 X Raero (where Raero is the maximum kinematic range of the missile) and is based on the limits of the ASEC and a 45 degress LOS to target limit. Blended aircraft and missile steering is provided for target ranges between 1.2 Raero and Raero. Inside Raero, the steering provides optimal horizontal and vertical missile steering. Once inside mid-point Rtr (half-way in the maneuver zone), the ASC provides Rmin steering. The pilot follows the ASC cue by rolling until the cue is on the HUD centerline above the center position of the ASEC and then pulls the aircraft (if commanded) to put the ASC in the center of the ASEC. When the target range is greater than Raero, a limit cross (X) is displayed inside the ASC to indicate that an AIM-120 shot does not exist even if the pilot performed a loft maneuver. The limit cross will also be displayed when the required lead angle exceeds 60 degrees, even if the target is nominally in range. Neither the ASEC nor ASC are displayed on the HUD in DGFT mode.

The following diagram sums ASC/ASEC relationship to the DLZ and the type of steering provided.



A/F Pole Cues – A-Pole (ownship to target range when AMRAAM goes active with MPRF) and F-Pole (ownship to target range at missile impact) ranges were added to the DLZ. A-Pole or F-Pole range indications, for the missile on the rail, were added to the DLZ below the Target Range Cue caret and Closure Rate indication. Indications are in NM with an M suffix for A-Pole (A is not used to avoid confusion with AMRAAM active indications) and an F suffix for F-Pole.

A-Pole or F-Pole range indications for the missile in-flight are added immediately below the DLZ. Post-launch cues are relative to winning Missile-Of-Interest (MOI, the missile aimed at the bugged target with the longest time remaining until impact) for the current bugged target. Changing the bug in TTS or TWS will provide A/F pole cues for the MOI for each if missiles are in flight to both.

Time remaining indication for the in-flight MOI is still displayed in the second field below the DLZ. This can show one of several prefixes: "A" (time to HPRF), "M" (time to MPRF), "T" (time to intercept) or "L" (time to termination; basically missile time of flight remaining). Note that this timer is more persistent than before. It's dynamic for bugged target and a winning MOI. For a winning missile where you lost track or snipped, the count becomes a simple stop watch. One note of caution: it's not a bug that the timer doesn't count down in linear fashion for a bugged target with an active DL missile. The time to intercept is calculated dynamically and takes target maneuver into account so it's perfectly reasonable for that count to *increase* even in some cases.

Missile datalink has been fixed to allow you to support up to six missiles in flight. These may be fired on up to two targets (TTS) or up to six targets (TWS). You can get A/F pole and time remaining data for all six missiles if you have six in flight and provided that their intended targets are still present on the FCR as track files.

HUD, HMS and FCR displays all include updated A/F pole cues.

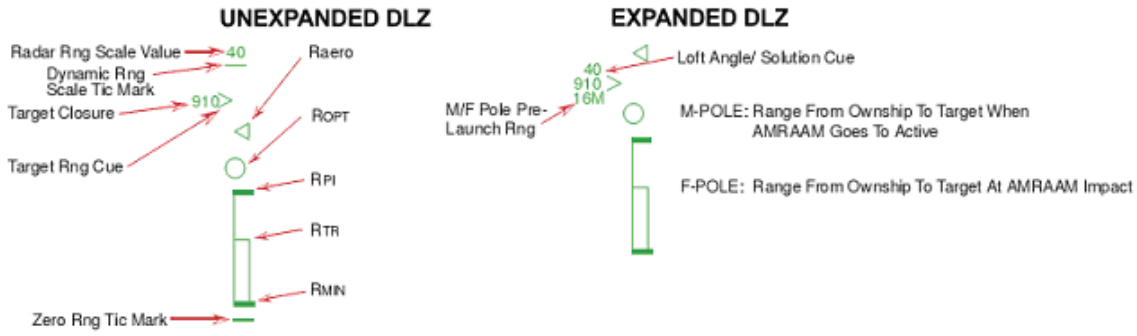
The radar model for AMRAAMs now includes HPRF (Husky) mode for favorable target geometries. The missile will activate the seeker well before the normal MPRF (Pitbull) range and attempt to track. HPRF is better at tracking high aspect targets with high closing range rate. Datalink guidance will continue up to MPRF unless the pilot commands a snip (drops the radar track) before that time. During HPRF with host DL guidance, the missile will use the best tracking solution available (either seeker or host DL guidance). HPRF and MPRF activation are now entirely based on range to target (used to be time-to-run based). Ranges coded are rough guesses for the AIM-120B. Other (active radar) missiles will need specific dat file edits (the AA-12 has been done already).

The variables are:

- AspectSelectorSize – HPRF requires target aspect to be within this many degrees of 180.
- HuskyMinClosure – HPRF will not activate unless range rate is higher than this threshold
- HighAspectBomberHusky – Select LARGE target (AIM120 SMS page) to use this as HPRF range gate
- HighAspectFighterHusky – Select MED or UNKN target (AIM120 SMS page) to use this as HPRF range gate
- BomberPitbull – Select LARGE target (AIM120 SMS page) to use this as MPRF range gate
- FighterPitbull – Select MED or UNKN target (AIM120 SMS page) to use this as MPRF range gate

Note: you can select SMALL in the SMS page. Doing so doesn't make a lot of sense in Falcon4 since this is apparently intended for targeting small RCS targets although it might be useful against helicopters. If you do choose small, MPRF ranges are reduced by around a third compared to MED.

Bug fixes relative to older versions of BMS in this general area: fixed bogus clearing of target when you flip between override modes or SMS changes from one missile type to another while you have a bugged target – now the lock is maintained. There is also a fix for datalink – previously last slammer was never provided with any datalink guidance making it significantly worse in performance compared to a missile fired with more of the same (slammer) type left on the rail.



AMRAAM DLZ Transitions

1) Unexpanded AMRAAM DLZ Scale: When the current target range is greater than 125% of Raero , the AMRAAM DLZ is displayed on the HUD in an unexpanded configuration. On the HUD the upper and lower range scale tics are displayed at their static locations and the radar range scale digital readout is displayed above the upper range scale tic. The remaining AMRAAM DLZ parameters (target range caret, Raero , Ropt , Rpi , Rtr , and Rmin) are positioned relative to the upper range scale tic which represents the current radar range scale value and the lower tic that represents zero range.

2) Expanded AMRAAM DLZ Scale: The AMRAAM DLZ transitions to the expanded scale when the target range is less than or equal to 125% of Raero. The upper and lower range scale tics and the radar range scale are removed from the HUD when the DLZ transitions to the expanded scale. The upper boundary of the expanded range scale on the HUD is dynamic and represents 125% of Raero. All DLZ parameters are positioned proportionally to the value of the range scale. However, as target range decreases below 125% of Raero, the Raero triangle is always displayed at 80% of the DLZ scale range.

AIM-9 DLZ Transitions

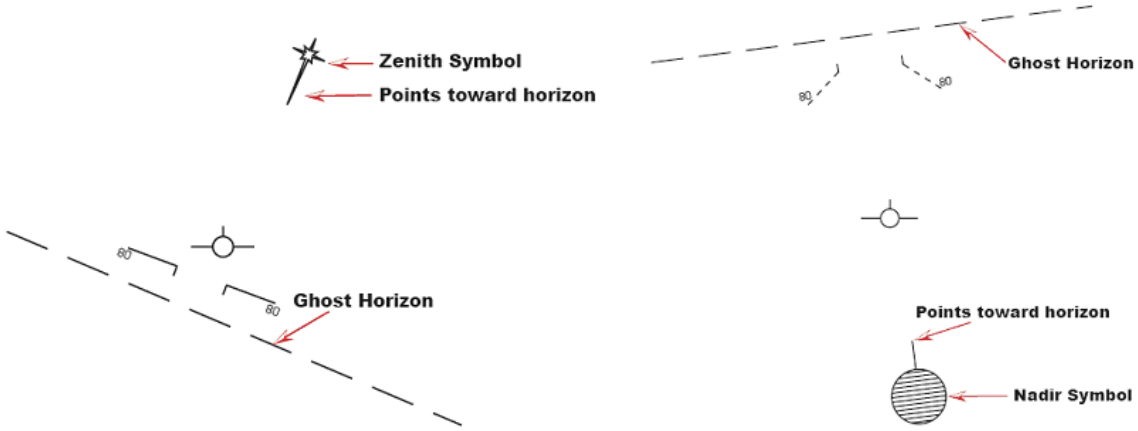
The AIM-9 DLZ also has unexpanded and expanded scales to maintain consistency with the AIM-120, except a value of 110% of Rmax1 is used.

Attitude Awareness Arc

Attitude awareness in DGFT is provided on the HUD through the Attitude Awareness Arc (AAA) and the Ghost Horizon Line (GHL). The Attitude Awareness Arc provides a global picture of pitch and roll attitude in the form of an arc. Roll is provided by the angle of an imaginary line through the ends of the arc in relation to the HUD. During wings level flight the arc gap is toward the top of the HUD, while during inverted flight, the arc gap is toward the bottom of the HUD and indicates the upright attitude. The arc represents the ground. The attitude arc is not adjusted for winds or aircraft yaw and is roll stabilized about the center of the HUD Total Field Of View (TFOV). The curvature of the arc represents the direction of the ground, and an imaginary line connecting the ends of the arc represents a line parallel to the horizon. In level flight the arc forms a half circle. As the aircraft increases nose-up pitch, the arc length decreases. As the aircraft noses over below the horizon, the arc length increases. The arc is at its smallest length at 87-degrees nose-up and is at its longest length at 87-degrees nose-down. The GHL appears when the aircraft exceeds +/-10 degrees of pitch to enhance attitude awareness in conjunction with the Attitude Awareness Arc.

Zenith/Nadir symbols

Addition of Zenith/Nadir symbols in the HUD. At +/–90-degrees pitch, the Zenith and Nadir symbols are displayed respectively, with arms pointing to the horizon. The Zenith symbol is an elongated star, the longest arm of which points to the horizon line. The Nadir is a 31.4-mr circle with a 15.7-mr line extending from the circle and pointing to the horizon. The Nadir has twelve lines inside the circle that parallel the horizon line to distinguish it from other similar symbology.



HUD Sampling

If you study enough HUD videos on Youtube, you'll notice there are small delays in numerical data sampling such as airspeed and altitude (either caused by HUD or CADC processing and/or the 'natural' delays in pitot-static systems) and not an instantaneous display of individual numbers. An example is a quick acceleration in airspeed will 'jump' from value to value...i.e., 102, 105, 109, 113. In previous versions of F4, the code would display every single digit through an acceleration/deceleration or changes in altitude. Now that data is sampled. The result is basically identical behavior to real life.

More studying also reveals a "lag" effect in the inertial system and can be clearly seen in the behavior of the aircraft pitch ladder and horizon line (it is slightly behind what the aircraft is actually doing). This effect has also been realistically reproduced. In addition, other symbology such as TD boxes, etc have been properly "lagged" to produce a more realistic HUD representation.

Stores Management System

The Stores Management System (SMS) in Falcon has received a complete overhaul. Much of this work will be somewhat transparent to the user, but the improvements are there nonetheless. Many of the improvements have been to make the code more consistent, efficient, cleaner, and more “friendly” in terms of playing nicely with other parts of the Falcon code. The following is what the end-user will see, when comparing to previous versions of BMS and Falcon.

A long-standing problem with Falcon and its SMS, was the fact that it was unrealistic in jettison procedures—both selective jettison and emergency jettison. Essentially, when jettisoning stores (weapons), the entire store/pylon assembly was jettisoned from the wings. In reality, the circumstances of what comes off during a jettison greatly depends on what weapon(s) are carried, whether or not the aircraft is carrying racks (launchers) or something like a TER (Triple ejector rack). The basic point is that the wings would essentially be empty of all pylons (excluding AA missile pylons), when the pylons (and the drag!) are actually bolted to the wings and can’t be jettisoned. BMS has rebuilt the SMS code to allow such differences amongst a wide-variety of stores and their individual differences. More information about the BMS rack data and how it’s built can found in the Appendix. Since Falcon is primarily an F-16 simulator, BMS focused only on the SMS of the F-16.

Selective Jettison (S-J)

The first thing the pilot will notice is some graphical differences. The line box drawn around the store/pylon(s) are gone, and now replaced by inverse labels, upon when a station's store/rack is selected for jettison. The S-J page and the S-J mastermode are selected by depressing OSB 11 adjacent to the S-J label on any SMS page. This will allow the pilot to jettison weapons and racks unarmed or unguided from selected aircraft stations. Only jettisonable stores will be displayed for selection. The pilot presses the OSB adjacent to the station displayed on the S-J page. The selected station's bottom-most store is highlighted on the S-J page, indicating that it is selected. If a jettisonable rack is also loaded, it may also be selected on the second depression of the OSB. A third depression will then deselect all stores on that station. The pilot can preselect a selective jettison configuration while in S-J mastermode, which will be remembered during mastermode transitions. The stores are jettisoned using the pickle button when the Master Arm switch is in ARM. After the stores are released, the highlighted stations are removed from the S-J page and the associated weapon quantity reads zero. The S-J mode also bypasses any other weapons settings.

Using the SMS in Combat

Also related to the SMS, is how it functions in combat. Previously, the SMS did not do a good job at remembering what weapons you called up for a particular mastermode. Now, it should remember the particular weapon type you select. This applies to all the mastermodes—Air-to-Air, Air-to-Ground, NAV, Dogfight, and Missile Override. What this means is the pilot can program the weapons he wants to come up first in a particular mastermode.

Hands-On Missile Select

Depress and Release (D&R) of the Missile Step switch on the side stick controller provides hands-on selection of a stores station. When in an A-A master mode, D&R of the Missile Step switch for <0.5 seconds selects the next stores station in the rotary with the selected weapon type. The selected station is identified on the SMS Base page by inverse video. Hands-off changes remain the same.

When in an A-A master mode, depress and hold the missile step switch >= 0.5 seconds to select the next missile type in the A-A weapon selection rotary. The avionics system automatically selects the next missile type in the A-A rotary and displays the newly selected weapon mnemonic adjacent to OSB #7 on the SMS Base page. The missile type (SRM, MRM) will also change on the HUD if the new missile is a different type.

Gun Related Changes

The gun option was taken out of the weapon rotary. You need to hit OSB 2 to get to the GUN in an AA mode or (easier) just use DOGFIGHT override. This prevents getting to the gun when using the missile step/missile type switch command and not being able to get back to missiles.

There was significant change to the implementation of the gun code.

- Guns now have longer reach.
- Guns have dispersion.
- The code can simulate every round or 1/10th of rounds.

A new config var is available to disable simulating every round: g_bOneTenthOfBullets (default false) will simulate 1/10th of the rounds fired. This is similar to the original code.

Bullet dispersion was also added, although this is currently semi hardcoded. It can be altered during run time by issuing the chat line command: .gundis (default .34). Values are specified in degrees, with the default being based on the F-16.

APG-68 Radar

CRM Range-While-Search Submode (RWS)

Two Target SAM

TTS simultaneously tracks two targets while maintaining a RWS scan volume centered on the secondary (non-bugged) target.

The following paragraphs provide an overview of CRM RWS operations:

- Designating on an RWS search target changes it to a bugged target and places the FCR in RWS SAM submode.

- Designating on a RWS search target while in SAM will transition the radar to TTS submode. When TTS is entered, the first bugged target remains the primary (currently bugged) target, the second designated target becomes the secondary target, and the scan volume is adjusted to permit dwells on each target as the radar continues to RWS. When both the bugged target and the secondary target are beyond 10 nm, the search volume is fixed at 50 degrees wide by three bars in elevation. The scan pattern is centered in azimuth on the acquisition cursor, and is controlled in elevation via the elevation trim control. TMS-right for less than 1 second will step the bug between the two TTS targets. When either target is within 10 NM, search is suspended and the radar spends full time tracking the bugged and secondary targets (ping pong). When the scan pattern is moved off of both targets in TTS, the scan width will adjust to optimize track versus search time. As the scan pattern is moved away from both targets, the scan width will become narrower. When the bugged target is within 3 NM, the radar automatically drops track on the secondary target and reverts to STT on the bugged target.

- Designating on a bugged SAM target places the radar into STT.

- TMS-forward actions change submodes from RWS to SAM to TTS to STT. TMS-aft changes the submodes back in the same sequence as TMS-forward. For example, if the pilot designates a target and puts the radar in the SAM submode, a subsequent TMS-aft will return the bugged target to a search target and the radar will revert back to CRM RWS submode. TMS-AFT in STT will place the radar in TTS if a secondary target is being extrapolated, or SAM if there is no secondary target. TMS-aft from TTS will place the radar in SAM, and TMS-aft from SAM will place the radar in CRM RWS.

- Depressing TMS-right for more than 1 second, from any CRM RWS mode with or without a tracked target (SAM, TTS, or STT), transitions the radar to TWS and retains any pre-existing bugged targets.

- Commanding TMS right and hold from TWS with a bugged target and system track files will transition the radar into two target SAM. The bugged target in TWS will become the primary (bugged) target in TTS. The TWS system track file to be carried through as the TTS secondary target is based on MMC and FCR prioritization, based on the following:
    - Cursor attach tank target or system track file
    - Target against which an AIM-120 has been launched (active data link) in descending order of time until active (not implemented yet)
    - Target range. (not implemented yet)

Track While Scan (TWS) / “Twiz”

Previously, the Target Acquisition Cursor attached itself to targets even with radar lock or designation broken, or gimbal limits exceeded. This has now been fixed.

The TWS radar mode has undergone a complete overhaul in how it is mechanized. It should operate much more like the real thing than ever before. This includes the number of targets it can track at once (which was previously unlimited), how it builds and maintains track files, how it dumps them and the azimuth and elevation limits of TWS.

Using TWS

To switch to the CRM-TWS (combined radar mode) sub-mode, the pilot may either hit OSB 2 three times or TMS right to switch from CRM-RWS to CRM-TWS. Upon entering TWS, the radar azimuth will initialize to an “A2” or 50 degree azimuth scan (25 degrees on either side of boresight) and the elevation will initialize to a 3-bar “3B” scan.

TWS Mechanization

As its name implies, TWS tracks multiple targets while searching for others. It is mechanized to begin forming track files (triangles, or hollow squares with a tic if you’re using EPAF radar cues) automatically from RWS search hits (solid squares) when the radar receives two hits (meaning the radar detects something twice) in 6.5 seconds. The radar is able to track 10 targets simultaneously. Since the radar does not pause on the track files while scanning, the track’s positions are extrapolated in between updates (when the radar detects them again). If a target is not updated, i.e., detected in 13 seconds, the radar will dump the track file until the target is detected again upon which it will rebuild it into a track file. A dump could happen for a number of reasons including a target moving out of the radar’s current azimuth scan, elevation scan, or both. Also for example, if the pilot is tracking 10 targets and decides to designate on a search target, the radar will dump the lowest priority track and automatically upgrade the search target into a track file. If the radar has not received a hit on a track on its return scan where the track should be (or rather, where the radar thinks it should be based on the target’s last heading and speed), the track file will turn from yellow to red to indicate this. When the track is detected again, it will turn back to yellow. If a track is no longer detected, it will turn red like previously mentioned and extrapolate for 13 seconds total. The last 5 seconds before the radar dumps the track, the track will begin to flash. Tracks are prioritized by range and the order in which they were built.

Three scan patterns are available in TWS. They are:

± 60, 2 bar

± 25, 3 bar

± 10, 4 bar

Without a bugged target, the azimuth scan centers on the cursors and elevation is controlled manually. When a target is bugged, the azimuth is biased to keep the bugged target in the scan and the elevation is centered on the bugged target. If the antenna elevation is tilted while the pilot has a bugged target, upon dropping the bug, the elevation scan will move according to what the pilot commanded to reflect the position set by the antenna elevation controls.

There are two ways to bug targets. The pilot may either slew the cursors over to a track file (or a search target) and designate or may TMS-right to bug the closest track file. Further TMS-right’s will step the bug to the next highest priority track file. The pilot may enter STT (Single Target Track) by slewing the cursors over the bug and pressing TMS-forward. This will erase all search targets and tracks from the radar, although the tracks will extrapolate for 13 seconds. If the pilot undesignates (TMS-aft) to return to TWS, the extrapolated tracks will reappear and the target will be bugged. If TMS-aft is commanded again, the pilot will drop the bug and the radar will continue to TWS. If TMS-aft is commanded a third time, the radar will dump all tracks and begin rebuilding tracks automatically. If TMS-aft is commanded a fourth time, the radar will go back into CRM-RWS.

Targets are detected in TWS as they are in RWS, and search detection symbols (small filled squares) are displayed at the target range and azimuth. Search detections may be either manually or automatically acquired and tracked. Search detections whose ground speed along the radar line of sight is greater than 200 knots will automatically be acquired and tracked by the radar as described above. Targets whose speed is slower than 200 knots can be acquired manually by placing the acquisition cursor over the target and designating. The speed gate prevents using a track slot for a low threat target or a potential false alarm.

Spotlight Scan

Depressing and holding the TMS to the designate (up or forward) position for longer than 1 second will command the radar to spotlight scan (±10 degrees azimuth by 4 bars elevation). Spotlight scan is initially centered about the acquisition cursor and antenna elevation knob setting and can be slewed. The radar attempts to acquire and track the target within the acquisition cursor at release of the designate position of the TMS.

Spotlight scan in TWS is similar to spotlight scan in RWS. The TWS spotlight search volume is initially centered about the cursor. The spotlight search volume is not biased by the TWS track files, but is controlled through the use of the cursor and the elevation thumbwheel. In other words, the pilot can override the TWS bug priority in azimuth and elevation. Spotlighting outside the TWS priority scan volume will help establish track files on groups outside the previous TWS scan volume. Track updates on the target of interest (TOI) and targets with AIM-120 missiles in flight will only occur if spotlight scan volume includes these targets.

Hot lines

Hot lines provide in a graphical format, the component of the targets' ground speed along the antenna line-of-sight. Hot lines pointing up the display indicate tail aspect targets, while hot lines that point down indicate potentially higher threat targets on a head aspect. Pilots need to be aware that near beam targets at 100" right/left aspect will show a head vector, or 80" right/left aspect will show a tail vector. Hot lines are shown for RWS search targets.

GM Snow Plow Mode

Air-to-Ground GM SnowPlow mode has received a significant overhaul to be more realistic. The following is a brief description of its changes and how it functions. Pay special attention on how to enable cursor slewing.

Depress OSB 8 next to the SP mnemonic to select the snowplow option. The mnemonic highlights indicating that you are in the SP mode. SP sighting directs each sensor line-of-sight straight ahead in azimuth, disregarding any selected steer points. In the GM, GMT, and SEA modes, the ground map cursor will be positioned at half the range selected, i.e., the center of the MFD. The cursors remain at this range while the ground map video moves, or "snowplows," across the MFD. At this point, there is no SOI, and the cursors cannot be slewed. The cursors can be slewed to a target or aim point with the CURSOR/ENABLE switch **after** you **ground stabilize** them by using **TMS forward**. TMS forward establishes the radar as the SOI and enables cursor slewing. TMS forward again over a target to command single target track. All cursor slews in SP are zeroed when SP is deselected. After ground stabilizing, the point under the cursors at the time of stabilization effectively becomes your steer point. All NAV and weapon delivery steering and symbology, including great circle steering, will be referenced to this "pseudo steer point." Displays return to the previously selected sighting point when SP is deselected. For example, SP can be used to accomplish an FCR mark on a point 5 nm in front of your position when the steer point selected is 40 nm away. It may often be used with IR Mavericks where target coordinates are not known in advance, for example. Another application of SP is for weather avoidance (not implemented).

The OSB 8 (SP) button on the GM FCR page is now a toggle to get in and out of SP mode. The pilot can also exit SP mode by switching to any A-A mode FCR page or by changing steer points (after you have ground stabilized SP only – pre-designate changes of steer point have no effect); or by selecting any A-G visual modes (CCIP, DTOS, STRAFE, EO-VIS). That leaves out only a couple of the more esoteric real conditions. There are also a couple of TMS changes associated with this. TMS aft will now only drop a ground target lock placing the cursor at the same point it was before a lock attempt was made; it will no longer clear slews and remove ground stabilization. You must use the CZ

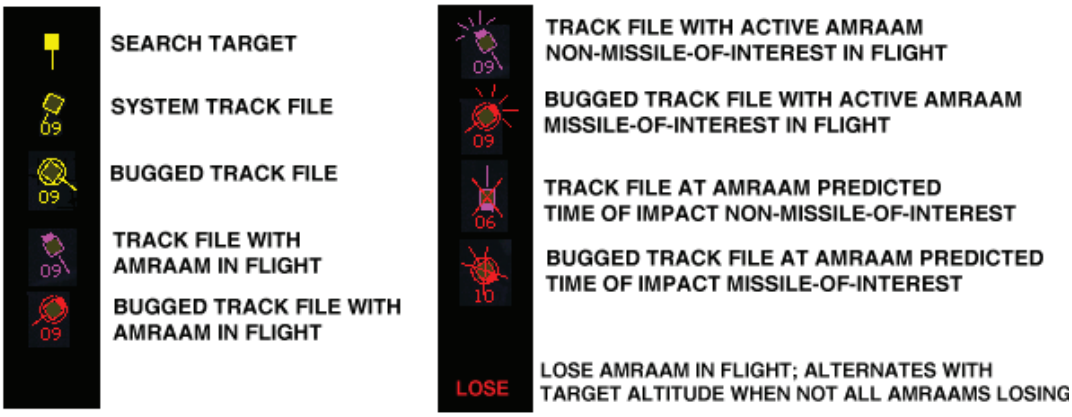
(OSB 9) to zero slews and clear ground stabilization now. These changes also fix certain ICP presses that would undo ground stabilization and reset SP mode unexpectedly.

Additionally, the GM range AUTO bumping function is revised in this version. It works more intuitively now with the range bumping up if the cursor is at 95% of the way up the MFD and bumping down if the cursor is at 42.5% of the way up the display or less. Note the bump will only happen if and when the cursor is not being slewed! As a final note, TTG (Time-to-go) has been added to the GM radar scope when you are in STP mode or SP mode with a ground stabilized aim point.

AMRAAM FCR Symbology

The symbology on the Bscope when an AMRAAM is in flight is now stored when the pilot bugs another contact. TWS mechanization: The pilot fires on a track and the normal AMRAAM symbology is displayed. The pilot may then bug another track or TMS right and fire on another track. The first track's symbology will be retained and the track will turn magenta in color. The pilot may bug additional tracks and shoot missiles. The AMRAAM missile timing information will be retained for all missiles in flight for their respective track files and shown for the current track.

Air-to-Air FCR Symbology



Other Radar Changes

Fixed the ACM slewable scan search altitude display numbers to show negative numbers and red when negative. Fixed the spacing between them so they're not scrunched up together. Make the BORE and SLEWABLE ACM crosses bigger in size - based on HUD videos. Also a fix for the slewable circle in the HUD to go to the edges of the slewable cross. Fix for altitude coverage numbers in Bscope. They should have been based off a 5 NM range, but instead were based off of the RWS/TWS cursor ranges which lead to erroneously large altitude coverages.
The bar scans (the number of bars in the scan, rather) was incorrect for both ACM 30x20 and ACM Slewable. 30x20 was corrected from a 5B to a 4B and Slewable was corrected from a 3B to a 4B.

DBS1/2 modes will now scan from outboard to inboard. For example if the target is 20 degrees off the left side of the nose, the antenna will sweep from left to right, then reset to the left and repeat the process. The variables to configure the DBS scan rate factors are no longer needed and have been removed from the Config Editor. The DBS scanning now very closely mimics real life. Also, if the pilot is in any of the expanded modes and then changes the range selection, the radar will come out of the expanded mode and change to the commanded range scale.

Various additions to the AG Freeze (FZ) mode—the lat/long of the cursors are now displayed on the upper right of the scope, and the target bearing/range are displayed in the lower left. Changing the range selection, the OSB next to FZ, or changing FOVs will exit FZ mode.

ACM-One Switch Action to NO-RAD:

1. Any ACM radiating submode, TMS aft enters ACM 30x20 NO-RAD.
2. From 30x20 NO-RAD, TMS aft enters ACM 10x60 radiating submode.

TMS aft is the only switch action that commands a break lock from an ACM track mode.

What this means is in any radiating ACM mode, whether BORE, 30x20, Slew, or 10X60, the TMS aft command will drop the current radar track if there is one and command 30X20 and snooze the radar. If TMS aft is commanded again, the radar will switch to 10X60 and radiate. A further TMS aft will again switch to 30X20 and snooze the radar. With a track in any of the ACM modes, switching to another mode (with the exception of 10X60) will not drop the current track, but only switch modes, i.e., in BORE mode→TMS right→30X20.

Cursor area of effect is now larger. Default 105% of the size of the captain's bars symbol "| |" as you see it on the screen this means that you get SAD display a little sooner than before as the cursor moves very close to but not quite over a blip on the display. This should greatly improve sorting capability because of the larger area of effect and the way the code now differentiates targets that are very closely drawn on the display. For example, sorting the trailer in a lead-trail group – Move the cursor up to the upper side of the group and TMS fwd. The larger cursor area will allow you designate the trailing contact without having to super-precisely put the cursor right over the contact.

Rewrote the logic for radar target history management. Formerly, anytime the target was painted and detected or extrapolated, the history state was updated, meaning that it was possible to get very large numbers of "history" entries in a single radar frame. This was the underlying cause of the blips dribbling down the MFD at speed when being extrapolated (painted but not detected lately). Now the history is only pushed down once per frame so multiple hits in one frame refine the top of the history stack updating the position data if it's changed (simulating improved track quality) but such updates don't overwrite history data from previous radar frames any longer. This makes display of history work much better.

Extrapolation code for SAM rewritten completely. Extrapolation is no longer applied to search hits in any mode (that was bogus apparently, especially in RWS which does no extrapolation at all now). An expert asserts that the actual radar computers are VERY good at extrapolating target position in three dimensions so the old scheme of just adjusting range only is gone in favor of near perfect (well OK, actual) position updates. The main difference now between added history and extrapolated is that extrapolated history doesn't update the last radar hit time so the track will start degrading if the radar doesn't pick it up again within some reasonable time frame (pattern dependent of course). If the radar has not detected the bug/track target lately (like in the last 5 paints where the beam passes over the target aircraft but doesn't get a strong enough return) and the radar is coasting (4 seconds since the last actual hit) extrapolation continues. After the coast time the symbol will flash to indicate imminent loss of track status; sometimes you won't even see this depending on where the target was in the scan pattern...either way the bug/track will shortly disappear unless you get a paint and detection almost immediately.

SAM mode entry antenna steering modified. Previously there was a call to ChangeMode() that resulted in the radar antenna being "homed" as part of the mode change from RWS to SAM. The time taken to home the antenna and then start scanning again was often taking just longer than the pattern time for SAM. This in turn caused the target to be lost for one frame by the radar resulting in the SAM bug disappearing right after being designated. To fix this now, the ChangeMode() code does NOT home the radar antenna for changes into SAM mode. Also, on transition from RWS to SAM (via TMS fwd on a track), the radar antenna shifts direct to the bugged target for the standard SAM mini-scan dwell time and only then resumes scanning pattern. This behavior is realistic. Combined with the removal of the homing of the antenna on mode switch, this appears to fix the problem with the prematurely disappearing bug symbol.

SAM AUTO independent azimuth and elevation implementation completed – previously the antenna elevation was always manually trimmed. The seeker center management was completely rearranged into one spot in the code to rationalize it as part of this fix. For some reason it was scattered all over the place and pretty hard to follow as a result. Upon ST (single target) SAM entry, the acquisition cursor is positioned over the SAM target and follows it. The azimuth of the scan pattern will move to stay centered around the SAM target until the ACQ cursor is moved. If the

acquisition cursor is slewed, auto AZ is terminated and the acquisition cursor no longer follows the tracked target. Search elevation is centered at the elevation selected by the ANT ELEV knob.

Several fixes to mode transitions for history marking such that you should see less "incorrect" symbol types for the mode you are entering (e.g. when entering SAM, you get a Bug symbol right away and leaving STT you get a bug not a track and then a bug etc.). In general the symbology should surprise you less as you change modes between RWS, SAM, TTS and STT.

Search history is now never drawn for track or bug symbols. It was previously but this effect was less obvious because the history insertion was broken and the colors for the history paints were so weak. Now those problems are fixed, incorrectly drawing history for track/bug would really stand out as wrong.

Fixed minor display bug where the SAD display for a blank target (yes, the real radar does this same thing, doing this is not in and of itself a bug) would be drawn in green: now it's the same color as a standard search hit (yellow at present).

Bug fix for SAM and TWS given the revised scheme for managing radar frame history. The old code here simply did bad things for SAM and TWS when trying to correlate paints to previous hits. It failed to check the actual position against the projected position and then did not take into account the time since the last history was recorded versus the time between the last histories used to project new position. New function added to re-factor the projection code so it can work for both SAM and TWS. Checks to see if the target is roughly where we'd expect him to be using radar history to project the current position (compared to actual position). For track-while-scan modes like SAM and TWS the radar may lose the track if the target returns don't correlate with the previous sets of history data.

Added EXP FOV to RWS.

The lower and upper numbers of the cursor altitude coverage show negative numbers. The numbers are drawn in red when they are negative.

Sniper Advanced Targeting Pod (ATP)

Background

The Targeting pod (TGP) has received a number of enhancements. The pod can track air-to-air (A-A) and air-to-ground (AG) targets. The slewing controls that interface with you the pilot have been rewritten and as a result controlling the TGP should be much smoother and intuitive. The TGP should no longer randomly jump to targets as in previous versions and tracking a target behind you does not “swap” the controls (i.e., to slew the TGP left, you had to move your cursors right). Tracking techniques include Point Track (based on a high-contrast edge – i.e., a vehicle like a tank), Area Track (which is a tracking mode for large area targets like buildings), and Computed Rates Track (based on the last known angular rate of change for the target). The TGP may operate in all mastermodes (NAV, A-A, A-G, DGFT & MSL OVRD). The TGP has two optical Fields of View (FOV), wide (+/-4 deg.), narrow (+/-1.0 deg.) and one electronic expansion (EXP) FOV (0.334x0.334) when selected (mimicking Sniper, but XR features are not implemented). The TGP has a laser ranger that can be fired at a target for very accurate ranging data.

Control of the TGP is available both hands-on and hands-off. For hands-on control the TGP must be the sensor of interest (SOI). The SOI is moved to the TGP via DMS Down (or AFT). Hands-off controls are via the MFDs TGP Base page, the A-A Mode Menu page and the A-G Mode Menu page. Selecting the TGP mnemonic on the Master Format Menu page will access the TGP Base page. The TGP OFF page is displayed if there is no power applied to the targeting pod. Power is applied to the targeting pod by placing the right Hard Point (HDPT) switch on the sensor power panel to the RIGHT HDPT position. When power is first applied, the TGP is placed in Standby (STBY) mode while the system cools and “NOT TIMED OUT” is displayed in the upper center of the TGP Base page. After the Forward Looking Infrared (FLIR) system has cooled, “NOT TIMED OUT” is removed from the display and the pod